

CLOX Hosting & Infrastructure

Production-Grade Sovereign Architecture & Scaling Economics

1. Executive Cost-Reduction & Optimization Strategy

To surpass legacy operations and high-cost digital marketplaces, CLOX leverages a decentralized, automated 4PL digital ecosystem. The fundamental hosting strategy shifts from elastic, usage-billed premium third-party APIs toward a Sovereign, High-Efficiency Infrastructure model combining AWS Fargate and Hetzner.

By migrating structural data validations, mapping, and optimization calculations onto self-hosted environments and free public data services, CLOX structurally decouples operational processing volume from linear cost increases, dropping variable transaction costs by up to 92% and maximizing platform margins.

2. Core Hosting & Computing Architecture

The overarching cloud environment relies on an AWS-centric infrastructure complementing Hetzner's bare-metal/cloud instances for rigorous cost control and predictable scaling.

Hetzner + AWS Fargate (Primary Compute)

- **Predictable Scaling:** Infrastructure scaling is governed by predictable, flat-rate server instances rather than highly variable API call metrics, diluting structural overhead toward zero per trip as transaction volumes grow.
- **Sovereign Routing Cluster:** The proprietary Geospatial & Routing Engine is self-hosted entirely on AWS Fargate or Hetzner.

Microservices & Backend Stack

- **Python (FastAPI / SQLAlchemy):** Powers the algorithmic spot pricing engine, the dynamic fuel surcharge (FSC) indexing, and strict programmatic safety lockout engines.
- **Node.js & Asynchronous WebSockets:** Required for live spatial telematics streaming. Persistent, two-way WebSockets handle live GPS coordinate streaming without blocking database operations during peak transaction periods.

3. In-House API Cost-Reduction Playbook

CLOX strategically isolates critical operations in-house to protect unit economics from elastic 3rd-party vendor pricing.

Operational Axis	Legacy / High-Cost Framework	CLOX Sovereign Low-Cost Alternative
Mapping & Matrix Routing	Google Maps Distance Matrix API (often exceeds \$8,000+ AUD/month at scale).	Self-Hosted Valhalla Engine: Run on AWS Fargate/ Hetzner with free OpenStreetMap (OSM) data. Handles heavy vehicle mass limitations, HazMat bypasses, and multi-stop optimization natively.
Geofencing & Proximity Alerts	Radar.com Pro Tier (~\$600 AUD/month).	PostgreSQL / PostGIS: Deep database integration executes spatial queries (e.g., ST_Contains, ST_DWithin) directly inside the backend to handle geofencing, triggering free waiting timers and 20-minute ETA alerts natively at zero variable cost.
Address Autocomplete	Google Places Autocomplete API (usage-based billing).	Pelias or Nominatim: Self-hosted geocoding engine ingesting the free, authoritative Australian G-NAF (Geocoded National Address File) dataset for cost-free, highly accurate local address lookups.
Real-Time Telematics	Premium positioning management layers with heavy server-side processing.	EMQX (MQTT Broker): Offloads polygon calculation vectors to the client-side device. Streams compact MQTT payloads exclusively at perimeter boundaries to keep server loads extremely low.
Corporate KYB Onboarding	Paid regulatory lookups (e.g., easyAML at \$1-\$3 per check) for basic ABN validation.	ABR Web Services API: Direct, cost-free Australian Business Number (ABN) checks via public authentication GUID to verify active statuses. easyAML is reserved only for tier-2 DVS ID checks.

4. Financial Infrastructure & Telematics Guardrails

Escrow & Clearing (The "Do Not Touch" Zone)

While routing, mapping, and telematics are brought in-house, structural financial ledgers and clearing must remain securely outsourced to maintain strict PCI-DSS compliance and eliminate platform bad debt.

- **Stripe Connect:** Facilitates Destination Charges and 100% upfront escrow holds securely linking capital to unique Job IDs.
- **Monoova NPP & PayTo API:** Direct settlement over Australia's New Payments Platform. Replaces 1.75%–2.9% gross credit card friction with a flat transaction fee (\$0.20 – \$0.50) per automated split sequence (15% / 10% / 5% / 70%).

WebRTC Communication

To eliminate variable cellular communication costs (like Twilio Proxy at ~\$1.15 AUD per masked number plus SMS usage fees), the architecture recommends an in-app messaging and VoIP calling feature utilizing WebRTC running over the existing WebSocket layer. This enforces ecosystem retention while driving communication overhead to zero.

Scaling Economics & Revenue Reinvestment

As the platform scales across Australian corridors, the economic blueprints specify that revenue drawn from the Super Admin's 15% global infrastructure share must be explicitly prioritized to expand the computing capacity of the self-hosted AWS Fargate/Hetzner Valhalla routing clusters.

Because CLOX pays for fixed server capacity rather than variable API hits, the structural overhead per trip dilutes toward zero as transaction volumes scale. This establishes an insurmountable unit economic advantage over competitors burdened by legacy, usage-billed middleware.